

CONNECTED PSEUDOSPECTRA OF A NONNORMAL TOEPLITZ PATH: THE OPEN HATANO–NELSON CHAIN*

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Abstract. For the nonnormal tridiagonal Toeplitz matrix $A_n(a) = \text{tridiag}(a, 0, 1)$, $0 < a < 1$, we determine the finite-size transition at which its Euclidean ε -pseudospectrum becomes connected. The least singular value of $(x + iy)I_n - A_n(a)$ is nondecreasing in $|y|$, so every pseudospectral component retracts onto an interval of the real axis and is therefore contractible. If γ_n is the largest least-singular-value barrier between adjacent open-boundary eigenvalues, then connectedness is equivalent to $\varepsilon > \gamma_n$. We prove the strict finite-size comparison $\gamma_{n+1} < \gamma_n$ and sharp-order bounds $\gamma_n \asymp_a (n+1)a^{n/2}$. Consequently, the first connected matrix size is also the eventual connectedness threshold, and, for fixed a , it equals $2(\log(1/\varepsilon) + \log\log(1/\varepsilon))/\log(1/a) + O_a(1)$ as ε tends to zero. In the open-boundary Hatano–Nelson model, the coefficient $2/\log(1/a)$ is the right-eigenmode amplitude skin depth, in lattice-spacing units. The threshold is the finite length at which a fixed perturbation budget overwhelms every inter-eigenvalue barrier: below it, disconnected components provide perturbation-stable spectral labels, whereas at and above it those labels belong to one worst-case uncertainty cloud.

Key words. pseudospectrum, nonnormal Toeplitz matrix, Hatano–Nelson model, non-Hermitian skin effect, least singular value, Chebyshev polynomial

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1. Introduction. For a matrix $A \in \mathbb{C}^{n \times n}$, the Euclidean ε -pseudospectrum is

$$(1.1) \quad \sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}.$$

Equivalently, it is the union of the spectra of $A + \Delta A$ over all unstructured complex perturbations $\Delta A \in \mathbb{C}^{n \times n}$ with $\|\Delta A\|_2 < \varepsilon$; see [26]. For nonnormal matrices, these sets reveal spectral sensitivity and resolvent growth that the eigenvalues alone can miss; see [24, 26]. Classical spectral limit sets for banded Toeplitz sections go back to Schmidt and Spitzer [22]. Finite Toeplitz matrices are a setting in which the distinction between spectra and pseudospectra can be exponentially large in the dimension; their finite-section resolvents and pseudospectra have been studied in [6, 7, 8, 21].

We study the real tridiagonal Toeplitz path

$$(1.2) \quad A_n(a) = \text{tridiag}(a, 0, 1) = \begin{bmatrix} 0 & 1 & & & \\ a & 0 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & & a & 0 \end{bmatrix}, \quad n \geq 2, \quad 0 < a < 1,$$

where the three arguments of tridiag denote the subdiagonal, diagonal, and super-diagonal, respectively. Closed-form spectral, conditioning, and pseudospectral results for general tridiagonal Toeplitz matrices are developed in [17]. Although $A_n(a)$ is diagonally similar to a symmetric Jacobi matrix, the similarity is not unitary, and hence it does not preserve the Euclidean pseudospectrum. Our question is finite-dimensional and topological: as n grows at fixed a and ε , can connectedness appear and then disappear, or is there a unique transition?

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The same family is the clean open-boundary Hatano–Nelson chain [12, 13]. With the site convention in which t_R moves a particle from j to $j + 1$ and t_L moves it from $j + 1$ to j , assume $0 < t_R < t_L$ and use the positive-sign one-particle representative

$$H_n^{\text{OBC}} = t_L A_n(a), \quad a = \frac{t_R}{t_L}.$$

The dimensionless logarithmic parametrization used in the proof is

$$h = -\frac{1}{2} \log a > 0, \quad a = e^{-2h}, \quad r = \sqrt{a} = e^{-h}.$$

The diagonal gauge

$$\mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1})$$

and the upper-shift matrix $V_n = \text{tridiag}(0, 0, 1)$ give the dimensionless similarity

$$A_n(a) = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1}.$$

Multiplying by t_L and using $t_L r = \sqrt{t_L t_R}$ recovers the usual physical imaginary-gauge formula for H_n^{OBC} . Thus the open-boundary eigenvalues are real, while the right eigenmodes have the envelope $e^{-h(j-1)}$ and accumulate at the left edge. This is the non-Hermitian skin effect [18, 27, 28]; see [29] for a review. Under periodic boundary conditions the Bloch curve is the ellipse

$$\mathcal{E}_{\text{PBC}}(k) = t_L e^{ik} + t_R e^{-ik} = (t_L + t_R) \cos k + i(t_L - t_R) \sin k.$$

Early work connecting Hatano–Nelson–type localization with pseudospectra treated random bidiagonal and Toeplitz models [25]. Direct pseudospectral interpretations of nonnormal boundary sensitivity appear in [19, 20]. Mathematical foundations for the skin effect were developed in [2], while spectra and pseudospectra of tridiagonal k -Toeplitz matrices were treated in [4, 5]. For asymmetric lattices, Kiorpelidis and Makris [15] observed that eigenvalue-centered pseudospectral clouds expand with the lattice size and merge, beyond a critical size denoted by $\tilde{N}$, into a single cloud. In their description the pseudospectral abscissa grows exponentially up to that crossover, whereas long-time dynamical amplification need not stop there. The present N_c is a rigorous realization of this cloud-merger scale for the clean open Hatano–Nelson chain with the Euclidean unstructured pseudospectrum: we identify the exact finite-size criterion, rule out re-entrant splitting, and determine the asymptotic scale. See also the preprint [23] for a closely related pseudospectral analysis of the open Hatano–Nelson chain.

For general matrices, the geometry and coalescence of pseudospectral components have been studied by variational and distance-to-defectivity methods [1, 9, 16]. The present work complements that theory by determining, for the clean scalar tridiagonal Toeplitz path, an exact finite-dimensional connectedness criterion, contractibility of every component, and a strict comparison of the merger barrier in consecutive dimensions.

Our contributions are as follows. First, we prove vertical monotonicity of the least singular value, which reduces planar topology to a real-axis barrier problem and shows, more strongly, that every pseudospectral component is contractible. Second, a signed persymmetric pencil and a folded Chebyshev parametrization give the strict comparison of the largest barrier in consecutive dimensions. Third, rectangular

factorizations give sharp-order two-sided bounds and the resulting Lambert- W finite-size law. Connectedness concerns the union over all *unstructured* perturbations: it does not imply that any single perturbation supplies eigenvalues throughout a connecting bridge, nor does it describe the attainable set for a prescribed structured perturbation family. Its physical content is instead the loss of a component-wise, perturbation-independent spectral label at the prescribed resolution. The transition is also not the onset of the skin effect, which is already present in the dimensionless gauge similarity above.

The paper is organized as follows. Section 2 states the main result. Vertical monotonicity is proved in section 3; the strict barrier comparison occupies section 4; and the topological criterion and quantitative bounds are established in section 5. Numerical illustrations and the physical interpretation are given in sections 6 and 7.

2. Model and main results. For $n \geq 2$ and $0 < a < 1$, let

$$A_n(a) = \text{tridiag}(a, 0, 1).$$

When a is fixed, we write A_n for $A_n(a)$. For $\varepsilon > 0$, write

$$\sigma_\varepsilon(A_n) = \{z \in \mathbb{C} : s_{\min}(zI_n - A_n) < \varepsilon\}.$$

Define

$$N_e(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_n) \text{ is connected for every } n \geq N\},$$

$$N_f(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_N) \text{ is connected}\}.$$

Throughout the remainder of the article, put

$$r = \sqrt{a}, \quad J_n = [\mathbf{1}_{\{i+j=n+1\}}]_{i,j=1}^n.$$

Thus $A_n = V_n + aV_n^T$. Throughout, $\mathbf{e}_j$ denotes a Euclidean coordinate vector of the ambient dimension, and $U_j(t)$ denotes the Chebyshev polynomial of the second kind, with $U_{-1} = 0$.

When the dimension is n , write $N = n + 1$ and define

$$(2.1) \quad I_n^{\text{sp}} = \left[-2r \cos \frac{\pi}{N}, 2r \cos \frac{\pi}{N}\right], \quad g_n(x) = s_{\min}(xI_n - A_n), \quad \gamma_n = \max_{x \in I_n^{\text{sp}}} g_n(x).$$

The explicit lower and upper barriers used below are

$$(2.2) \quad \mathcal{U}_n(a) = r^n \csc \frac{\pi}{N},$$

$$(2.3) \quad \mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \frac{r^n(1 - r^2)}{(1 - r^{2N}) \sin \frac{3\pi}{2N}}.$$

THEOREM 2.1 (Main theorem). *For every $0 < a < 1$ and $\varepsilon > 0$, the following assertions hold.*

1. *For every $n \geq 2$, every connected component of $\sigma_\varepsilon(A_n(a))$ is contractible.*
2. *For $n \geq 2$, the barrier sequence is strictly decreasing, $\gamma_{n+1} < \gamma_n$, tends to zero, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n.$$

3. *The first-hit and eventual connectedness thresholds coincide:*

$$N_c(a, \varepsilon) := N_f(a, \varepsilon) = N_e(a, \varepsilon).$$

Moreover,

$$(2.4) \quad 1 + \max\{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\} \leq N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\},$$

where the maximum of the empty set is understood to be 1.

4. For fixed a , as $\varepsilon \downarrow 0$,

$$(2.5) \quad N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1),$$

and $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$.

3. Vertical monotonicity. We first prove the vertical monotonicity that will be used in the topological part of the argument.

LEMMA 3.1 (Vertical monotonicity). For every real x ,

$$y \longmapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

is nondecreasing.

Proof. Put $Y = y^2$, and, for $t \geq 0$, define

$$\mathcal{P}_n(Y, t) = ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \quad F_n(Y, t) = \det \mathcal{P}_n(Y, t).$$

The same definition is used in every dimension: for $k \geq 1$, $\mathcal{P}_k(Y, t)$ is obtained by replacing n by k , $F_k(Y, t) = \det \mathcal{P}_k(Y, t)$, and $F_0 = 1$. We shall prove the finite-dimensional implication

$$(3.1) \quad \mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

At a fixed value of Y , write

$$d = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For $n \geq 2$, the matrix $\mathcal{P}_n(Y, t)$ has diagonal

$$d - 1, d, \dots, d, d - a^2,$$

whereas for $n = 1$ its sole entry is $d - 1 - a^2$. In all dimensions its first upper diagonal is b , its first lower diagonal is $\bar{b}$, and its second off-diagonals are a . Let T_k be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal d and these two off-diagonals. Set $T_{-1} = 0$, $T_0 = 1$, and put

$$(3.2) \quad D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}.$$

Indeed, a determinant is affine in each diagonal entry. Replacing the first diagonal entry d by $d - 1$ subtracts the cofactor T_{k-1} , whereas replacing the last one by $d - a^2$ subtracts $a^2 T_{k-1}$. Thus, for $k < n$, D_k and R_k are, respectively, the leading and trailing principal minors of $\mathcal{P}_n(Y, t)$.

We record the continuant calculation that contains all the dependence on Y . With $T_k = 0$ for $k < 0$, a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every $k \geq 1$

$$(3.3) \quad \begin{aligned} T_k &= dT_{k-1} - (|b|^2 - a^2)T_{k-2} - (2a^2d - a(b^2 + \bar{b}^2))T_{k-3} \\ &\quad - a^2(|b|^2 - a^2)T_{k-4} + a^4dT_{k-5} - a^6T_{k-6} - a^2\mathbf{1}_{\{k=2\}}. \end{aligned}$$

For example, $T_1 = d$ and $T_2 = d^2 - |b|^2$; these also verify the only inhomogeneous coefficient in (3.3). Multiplying the recurrence by z^k and summing gives

$$(3.4) \quad \sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)},$$

where

$$Q(z, Y) = 1 - dz + (|b|^2 - a^2)z^2 + (2a^2d - a(b^2 + \bar{b}^2))z^3 + a^2(|b|^2 - a^2)z^4 - a^4dz^5 + a^6z^6.$$

Consequently, with $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$,

$$(3.5) \quad \mathcal{D}(z, Y) = \frac{(1 - z)(1 - a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1 - a^2 z) \mathcal{D}(z, Y).$$

For the second identity, first change the initial diagonal, producing D_k , and then change the final diagonal; its cofactor is D_{k-1} . Thus $F_k = D_k - a^2 D_{k-1}$, which proves the asserted generating function. Direct substitution of d and b gives

$$\partial_Y Q = -z(1 - z)(1 + az)^2(1 - a^2 z).$$

Differentiating (3.5) therefore yields

$$(3.6) \quad \sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1 + az)(1 - a^2 z)^2}{1 - az}.$$

Suppose that $\mathcal{P}_n(Y, t) > 0$. Then

$$(3.7) \quad D_k > 0, \quad R_k > 0 \quad (0 \leq k < n).$$

Put $u_k = \sum_{j=0}^k D_j D_{k-j}$, with $u_k = 0$ for $k < 0$. The coefficients of φ_a , beginning with the coefficient of z , are

$$1, \quad 2a(1 - a), \quad a^2(a^2 - 4a + 2), \quad 2a^{k-1}(1 - a)^2 \quad (k \geq 4).$$

Thus

$$(3.8) \quad \begin{aligned} \partial_Y F_n &= u_{n-1} + 2a(1 - a)u_{n-2} + a^2(a^2 - 4a + 2)u_{n-3} \\ &\quad + 2(1 - a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned}$$

Only the third coefficient can be negative. We now give the cofactor inequality that absorbs it. Let $j, k \geq 1$, $j + k = n - 1$. Deleting index $k + 1$ from $\mathcal{P}_n$ leaves blocks of sizes k and j , coupled only by the second off-diagonal a . In block form, the deleted principal submatrix is

$$(3.9) \quad \begin{bmatrix} L_k & a \mathbf{e}_k \mathbf{e}_1^T \\ a \mathbf{e}_1 \mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$(3.9) \quad D_k R_j - a^2 D_{k-1} R_{j-1} > 0.$$

Deleting index $j + 1$ instead gives the same inequality with j, k interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since $T_k = \sum_{\ell=0}^k D_\ell$, (3.2) gives

$$(3.10) \quad R_k = D_k + (1 - a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}.$$

Hence $\vartheta_k \leq a^2$ implies $\varpi_k \leq 1$. On dividing (3.9) and its companion by positive factors, we obtain

$$(3.11) \quad \vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2.$$

If both ϑ_j and ϑ_k were at most a^2 , then both ϖ_j and ϖ_k would be at most 1, contradicting (3.11). Hence, after interchanging j, k if necessary, $\vartheta_j > a^2$. Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals $(1 - a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$. Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (3.11) now gives $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$, and therefore

$$(3.12) \quad \vartheta_j \vartheta_k > a^4.$$

Consequently, for $n \geq 3$, applying (3.12) to $j = p + 1, k = n - 2 - p$ gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$(3.13) \quad u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}.$$

If the third coefficient in (3.8) is negative, write it as $-\mathcal{B}_{\text{neg}}$, where $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$. Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (3.8) and (3.13) prove (3.1). If that coefficient is nonnegative, the implication is immediate. For $n = 1$, one has $\partial_Y F_1 = 1$; for $n = 2$, one has $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$. Thus (3.1) holds in every dimension.

Finally fix $Y_0 \geq 0$. If the least singular value at Y_0 is zero, the desired inequality is automatic. Otherwise take $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$. On the maximal interval to the right of Y_0 on which $\mathcal{P}_n(Y, t) > 0$, implication (3.1) says that its determinant is strictly increasing. By continuity of Hermitian eigenvalues, a finite right endpoint would be a first loss of positive definiteness at a positive-semidefinite singular matrix, and hence would have determinant zero. This is impossible because the determinant has increased from its positive value at Y_0 . Thus $\mathcal{P}_n(Y, t) > 0$ for every $Y \geq Y_0$. Letting t increase to the least singular value at Y_0 proves the lemma. $\square$

4. Strict finite-size comparison of real-gap barriers. We next establish the strict comparison between consecutive dimensions. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity gives existence of every maximum, and the spectral endpoints have height zero. Every interior gap point lies outside the spectrum, so $xI_n - A_n$ is nonsingular there and $g_n(x) > 0$. Consequently every gap height is positive and every point at which it is attained lies in the open gap. Hence

$$(4.1) \quad \gamma_n = \max_{1 \leq k < n} \gamma_{n,k}.$$

LEMMA 4.1 (Strict barrier comparison). *For every $n \geq 2$,*

$$\gamma_{n+1} < \gamma_n.$$

We give the complete finite-path calculation. Persymmetry gives $A_n J_n = J_n A_n^T$, so $B_n(x) = (xI_n - A_n)J_n$ is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since J_n is orthogonal, the absolute eigenvalues of $B_n(x)$ are exactly the singular values of $xI_n - A_n$, including multiplicity. Consequently,

$$(4.2) \quad s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}.$$

Let $q_n(x, s) = \det(xI_n - A_n - sJ_n)$.

LEMMA 4.2 (The middle signed branch). *Write the eigenvalues of the real symmetric matrix*

$$B_n(x) = (xI_n - A_n)J_n$$

in decreasing order as $\alpha_1(x) \geq \dots \geq \alpha_n(x)$, and abbreviate $\lambda_k = \lambda_{n,k}$. If $\lambda_{j+1} < x < \lambda_j$, then

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \alpha_m(x), & n = 2m, \\ \alpha_{m+1}(x), & n = 2m + 1. \end{cases}$$

Proof. Let $A_n^{\text{sym}} = r(V_n + V_n^T)$. Since $A_n = \mathcal{G}_n(r)A_n^{\text{sym}}\mathcal{G}_n(r)^{-1}$ and $\mathcal{G}_n(r)^{-1}J_n = r^{-(n-1)}J_n\mathcal{G}_n(r)$,

$$(4.3) \quad B_n(x) = r^{-(n-1)}\mathcal{G}_n(r)(xI_n - A_n^{\text{sym}})J_n\mathcal{G}_n(r).$$

The sine eigenvector of A_n^{sym} belonging to λ_k is an eigenvector of J_n with eigenvalue $(-1)^{k+1}$. Hence (4.3) and Sylvester's law of inertia show that $B_n(x)$ has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

Counting signs gives

$$(4.4) \quad \nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even}, \\ m-1, & j \text{ odd}, \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even}, \\ m, & j \text{ odd}. \end{cases}$$

First let $n = 2m$. Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

In these coordinates

$$(4.5) \quad \begin{aligned} B_{2m}(x) &= \hat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \hat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned}$$

where $P_m(x) = xI_m - A_m$. Thus $\mathcal{R}_{\text{bd}} \geq 0$ and $\text{rank } \mathcal{R}_{\text{bd}} = 2$. For $\Gamma = \text{diag}(I_m, -I_m)$,

$$\tilde{B} = -\Gamma B_{2m}(x) \Gamma = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If $\tilde{\alpha}_1 \geq \dots \geq \tilde{\alpha}_{2m}$ are the eigenvalues of $\tilde{B}$, then

$$(4.6) \quad \tilde{\alpha}_k = -\alpha_{2m+1-k}.$$

Positive-semidefinite rank-two interlacing gives

$$(4.7) \quad \tilde{\alpha}_k \geq \alpha_k \geq \tilde{\alpha}_{k+2}, \quad 1 \leq k \leq 2m-2.$$

If j is even, the only candidates for minimum modulus have moduli α_m and $-\alpha_{m+1} = \tilde{\alpha}_m$, and $\alpha_m \leq \tilde{\alpha}_m$. If j is odd and $m \geq 2$, the two candidates have moduli α_{m-1} and $-\alpha_m = \tilde{\alpha}_{m+1}$; taking $k = m-1$ in (4.7) gives $\alpha_{m-1} \geq \tilde{\alpha}_{m+1}$. For $m = 1$, both eigenvalues in the sole gap are negative, so the larger one α_1 has minimum modulus. Thus α_m always attains the minimum modulus, and its sign follows from (4.4). Equality in an interlacing comparison is harmless: it only gives a multiple least singular value.

Now let $n = 2m+1$. Group the first and last m coordinates as above and leave the middle coordinate last. Then

$$(4.8) \quad B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}.$$

In an open spectral gap of A_{2m+1} , $P_m(x)$ is nonsingular. Indeed, the explicit path spectrum gives

$$\text{spec}(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \{ \lambda_{2m+1, 2\ell} : 1 \leq \ell \leq m \},$$

so its zeros are exactly the even-indexed eigenvalues of A_{2m+1} , none of which lies in an open spectral gap. The eigenvalues of the principal submatrix $\hat{P}_m(x)$ are consequently

$$s_1(P_m) \geq \dots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \dots \geq -s_1(P_m).$$

273 Cauchy interlacing gives

$$274 \quad \alpha_m \geq s_m(P_m) \geq \alpha_{m+1} \geq -s_m(P_m) \geq \alpha_{m+2}.$$

275 If j is even, (4.4) and interlacing give

$$276 \quad 0 < \alpha_{m+1} \leq s_m(P_m) \leq -\alpha_{m+2};$$

277 if j is odd, they give

$$278 \quad 0 < -\alpha_{m+1} \leq s_m(P_m) \leq \alpha_m.$$

279 Every other eigenvalue is farther from zero. Hence α_{m+1} attains the minimum modulus,
 280 with sign prescribed by (4.4); equality again only means multiplicity. At a spectral
 281 endpoint the zero is simple because A_n is similar to the Jacobi matrix A_n^{sym} , whose
 282 eigenvalues are simple. Thus the selected ordered branch extends continuously to zero
 283 at both ends of the closed gap. Finally, multiplication by J_n does not change singular
 284 values, which proves the assertion. $\square$

285 *Proof of Lemma 4.1. Folded continuants.* Put

$$286 \quad C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1+a^2+(1+a)s}{2a},$$

287 and define the folded discriminant and recurrence coefficient by

$$288 \quad \Delta_{\text{fold}} = ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2),$$

$$289 \quad \mathfrak{b} = (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

291 The global recurrence polynomial is
 (4.9)

$$292 \quad \left. \begin{aligned} 2a(u+v) &= C_0, \\ a^2(4uv+2) &= \mathfrak{b} \end{aligned} \right\} \implies \Pi(z) = (1-2auz+a^2z^2)(1-2avz+a^2z^2), \quad u, v \in \mathbb{R}.$$

293 On a real sheet with $\Delta_{\text{fold}} \geq 0$, we additionally put

$$294 \quad \xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

295 Then the pair $(u, v) = (\xi_+, \xi_-)$ satisfies the two symmetric relations in (4.9):

$$296 \quad \xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

297 The proof uses the root presentation only on these real sheets; the global transfer and
 298 generating identities use the polynomial Π itself. Here is a direct continuant derivation
 299 of the required folded formulas. Pair the coordinates in the order $(1, n), (2, n-1), \dots$
 300 In this order the symmetric matrix $(xI_n - A_n)J_n - sI_n$ is block tridiagonal. Its interior
 301 diagonal and off-diagonal blocks are

$$302 \quad D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block E_{off}^T . For even order the terminal block is

$$D_{\text{term}}^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix};$$

for odd order the terminal block is the scalar $x-s$, coupled to the preceding block by the column $(-1, -a)^T$. Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by $\det J_n$, is $q_n(x, s)$. Set $\mathcal{Q}_m^e = q_{2m}(x, s)$, $\mathcal{Q}_m^o = q_{2m+1}(x, s)$. We include the continuant verification to fix all boundary signs. We now give a complete minor transfer for the folded continuant. For $\tau \in \{e, o\}$, let K_m^τ denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m + 1,$$

after simultaneously permuting its rows and columns into the paired order. Thus K_m^e has order $2m$, K_m^o has order $2m + 1$, and

$$K_1^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^o = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For $m \geq 1$, increasing m by one adds one interior block at the exposed end:

$$(4.10) \quad K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = [E_{\text{off}} \quad 0],$$

where the zero block has the width required by K_m^τ .

For a matrix whose first two coordinates are exposed, write $K(\widehat{I} \mid \widehat{J})$ for the matrix obtained by deleting rows I and columns J . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value $\mathfrak{c}_m$, because K_m is symmetric.

Here is the one-step calculation. On the dense set on which both K_m and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in K_m , and let δ be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

331 taking the Schur complement of K_m in (4.10) gives

$$\begin{aligned}
 & \mathfrak{d}_{m+1} = (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\
 & \mathfrak{p}_{m+1} = -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\
 & \mathfrak{q}_{m+1} = -s\mathfrak{d}_m - \mathfrak{q}_m, \\
 & \mathfrak{c}_{m+1} = x\mathfrak{d}_m + a\mathfrak{c}_m, \\
 & \mathfrak{r}_{m+1} = \mathfrak{d}_m.
 \end{aligned}
 \tag{4.11}$$

333 Both sides are polynomial in x and s , and the nonsingular set used above is dense.
 334 The polynomial identity theorem therefore extends all five identities to the singular
 335 cases as well; no inverse is assumed in (4.11).

336 The five quantities in (4.11) satisfy one linear invariant. Define

$$337 \quad \ell_m = (1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m.$$

338 Direct substitution into (4.11) gives

$$339 \quad \ell_{m+1} = -a\ell_m.$$

340 For the two terminal blocks, direct evaluation gives

$$\begin{aligned}
 & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\
 & = ((1 + s)(a + s) - x^2, -a - s, -1 - s, x, 1), \\
 & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\
 & = (x(1 + a^2) + 2as - C_0(x - s), \\
 & \quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x - s).
 \end{aligned}$$

342 Thus $\ell_1 = 0$ in both parity classes, and consequently

$$343 \quad (4.12) \quad (1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m = 0 \quad (m \geq 1).$$

344 Because

$$345 \quad \det J_{2m} = \det J_{2m+1} = (-1)^m,$$

346 the first signed minor below is precisely the required determinant:

$$347 \quad \boldsymbol{\eta}_m^\tau = (-1)^m \begin{bmatrix} \mathfrak{d}_m^\tau & \mathfrak{p}_m^\tau & \mathfrak{q}_m^\tau & \mathfrak{c}_m^\tau \end{bmatrix}^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

348 Eliminating $\mathfrak{r}_m$ with (4.12) reduces (4.11) to the four-dimensional transfer

$$\begin{aligned}
 & \boldsymbol{\eta}_{m+1}^\tau = \mathbf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathbf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1 - a} & -\frac{s}{1 - a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}.
 \end{aligned}
 \tag{4.13}$$

350 The terminal conditions can be incorporated at $m = 0$ by setting

$$351 \quad (4.14) \quad \boldsymbol{\eta}_0^e = [1 \quad a^{-1} \quad 1 \quad 0]^T, \quad \boldsymbol{\eta}_0^o = [x - s \quad 1 \quad 1 \quad -1]^T.$$

Indeed, direct multiplication in (4.13) gives

$$\mathbf{T}_{\text{fold}} \boldsymbol{\eta}_0^e = \begin{bmatrix} x^2 - (1+s)(a+s) \\ a+s \\ 1+s \\ -x \end{bmatrix},$$

and

$$\mathbf{T}_{\text{fold}} \boldsymbol{\eta}_0^o = \begin{bmatrix} C_0(x-s) - x(1+a^2) - 2as \\ a^2 - s^2 + sx \\ 1 - s^2 + sx \\ a + sx - x^2 \end{bmatrix},$$

which are exactly the signed terminal minors above. Hence $\boldsymbol{\eta}_m^\tau = \mathbf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$ for every $m \geq 0$.

A direct determinant calculation gives

$$(4.15) \quad \begin{aligned} \det(\lambda I_4 - \mathbf{T}_{\text{fold}}) &= \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} &= (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned}$$

Cayley–Hamilton, followed by multiplication by $\mathbf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$ and extraction of the first coordinate, therefore yields, for $Z_m = \mathcal{Q}_m^e$ and for $Z_m = \mathcal{Q}_m^o$,

$$(4.16) \quad Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4.$$

Finally, applying $\mathbf{T}_{\text{fold}}$ three times to (4.14) gives the boundary residuals

$$\begin{aligned} \text{Res}_0 &= Z_0, & \text{Res}_1 &= Z_1 - C_0 Z_0, & \text{Res}_2 &= Z_2 - C_0 Z_1 + \mathfrak{b} Z_0, \\ \text{Res}_3 &= Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0, \end{aligned}$$

namely

	Res ₀	Res ₁	Res ₂	Res ₃
$Z_m = \mathcal{Q}_m^e$	1	$a(1 - 2\zeta_s)$	$a^2(1 - 2\zeta_s)$	a^3
$Z_m = \mathcal{Q}_m^o$	$x - s$	$-[x(1 + a^2) + 2as]$	$a^2(x - s)$	0

as asserted. Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as $(1+az)(1-2a\zeta_s z + a^2 z^2)$; for the odd sequence it is $(x-s) - [x(1+a^2) + 2as]z + a^2(x-s)z^2$. This proves (4.16). Since $\Pi(0) = 1$, it is invertible in $\mathbb{R}[[z]]$, and the two generating functions are the literal formal-power-series identities

$$(4.17) \quad \sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1+az)(1-2a\zeta_s z + a^2 z^2) \Pi(z)^{-1},$$

and

$$(4.18) \quad \sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x-s) - [x(1+a^2) + 2as]z + a^2(x-s)z^2) \Pi(z)^{-1}.$$

378 Now let $U_{-1} = 0$. The Chebyshev generating function gives

$$379 \quad \sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u))\omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

380 and, with $\mathbf{B}_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$,

$$381 \quad \sum_{m \geq 0} ((x - s)u - \mathbf{B}_s^{\text{fold}})U_m(u)\omega_{\text{gf}}^m = \frac{(x - s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

382 For a differentiable real function F , write

$$383 \quad [F]_{u,v} := \begin{cases} \frac{F(u) - F(v)}{u - v}, & u \neq v, \\ F'(u), & u = v. \end{cases}$$

384 Let $u, v \in \mathbb{R}$ satisfy the two symmetric relations in (4.9). Taking the corresponding
385 divided difference of the Chebyshev generating identities and putting $\omega_{\text{gf}} = az$ recovers
386 (4.17) and (4.18). Coefficient comparison gives

$$387 \quad (4.19) \quad q_{2m}(x, s) = a^m [\mathbf{f}_m]_{u,v}, \quad \mathbf{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)),$$

388 and

$$389 \quad (4.20) \quad q_{2m+1}(x, s) = a^m [\mathbf{g}_m]_{u,v}, \quad \mathbf{g}_m(t) = \left((x - s)t - \frac{x(1 + a^2) + 2as}{2a} \right) U_m(t).$$

390 The determinant recurrence and generating identities are global. The real-pair formulas
391 (4.19)–(4.20) are used only after the two symmetric relations have been verified; their
392 collision value is the displayed derivative. No complex square-root choice is made when
393 $\Delta_{\text{fold}} < 0$, and none is needed below.

394 *Half-angle chord parametrization.* Recall that $a = e^{-2h}$, and put

$$395 \quad \mathbf{s}_h = \sinh h, \quad \mathbf{c}_h = \cosh h.$$

396 For $K \geq 2$ and $0 \leq \theta \leq \pi$, define

$$397 \quad (4.21) \quad \Phi_K(\theta) = \sqrt{\mathbf{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta},$$

398 where the quotient is interpreted continuously at $\theta = 0, \pi$, and

$$399 \quad \Psi_K(\eta) = \sqrt{\mathbf{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h,$$

400 with the limiting value at $\eta = 0$. We spell out the half-angle reduction, including its
401 sign. Put

$$402 \quad y = \cos \theta, \quad z = \cosh \eta, \quad \mathbf{p}_\theta = \sqrt{\mathbf{c}_h^2 - y^2}, \quad \mathbf{p}_\eta = \sqrt{\mathbf{c}_h^2 - z^2},$$

403 and let $L_0 = 2r/\mathbf{c}_h$. If $x = L_0 yz$, $s = L_0 \mathbf{p}_\theta \mathbf{p}_\eta > 0$, and $\tau = \varsigma s$ with $\varsigma \in \{-1, 1\}$, then
404 direct expansion gives

$$405 \quad (4.22) \quad \begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & & v = 2y^2 - 1, & & \zeta_\tau = 2\mathbf{c}_h^2 - 1 + 2\varsigma \mathbf{p}_\theta \mathbf{p}_\eta. \end{aligned}$$

The real pair (u, v) satisfies the two symmetric relations in (4.9). For $K = 2m + 1$, use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for $K = 2m + 2$, use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (4.19) and (4.20), respectively, reduces either equation $q_{K-1}(x, \tau) = 0$ to

$$(4.23) \quad \mathfrak{p}_\theta U_{K-1}(y) = \varsigma \mathfrak{p}_\eta U_{K-1}(z).$$

In the odd-fold case the reduction also contains the factor $\mathfrak{p}_\eta y + \varsigma \mathfrak{p}_\theta z$. On a distinct-variable sheet this factor cannot vanish: after squaring, its vanishing would give

$$(\mathfrak{c}_h^2 - z^2)y^2 = (\mathfrak{c}_h^2 - y^2)z^2,$$

and hence $y^2 = z^2$. Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe $U_{K-1}(z) > 0$, whereas, for $j\pi/K < \theta < (j+1)\pi/K$,

$$\operatorname{sgn}(U_{K-1}(\cos \theta)) = \operatorname{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover, $\mathfrak{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$ and $\mathfrak{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$. Thus (4.23) is equivalent to $\varsigma = (-1)^j$ and

$$(4.24) \quad \Phi_K(\theta) = \Psi_K(\eta),$$

with

$$(4.25) \quad s = \mathfrak{s}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathfrak{c}_h} \cos \theta \cosh \eta.$$

Consequently $q_{K-1}(x, (-1)^j s) = 0$. Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of x , the preceding steps are reversible, and the two chord relations are equivalent to the corresponding signed equation. The determinant identities themselves extend to a folded-variable collision by the continuous divided-difference convention, but we invoke the chord equivalence only for distinct variables; the selected middle branch is proved below to be collision-free. Here $K - 1$ is the matrix size. If the exterior argument has crossed through 1, write it as $z = \cos \phi$, $0 \leq \phi < \pi/K$. Then the chord equation is $\Phi_K(\theta) = \Phi_K(\phi)$, and the second factor in the formula for s is $\mathfrak{s}_h^2 + \sin^2 \phi$. We denote the resulting value by

$$\mathfrak{s}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

Branch classification and exhaustion. We next identify exactly which of these chords is the middle branch of Lemma 4.2. Set

$$(4.26) \quad \mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h.$$

At $u = \pm \mathfrak{c}_h$, this denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$, then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{\mathfrak{c}_h^2 - u^2},$$

and its derivative is

$$(4.27) \quad - \sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{\mathfrak{c}_h^2 + u^2}{(\mathfrak{c}_h^2 - u^2)^2} < 0.$$

At each finite nodal endpoint the logarithmic derivative tends to the appropriate signed infinity, and it tends to $-\infty$ as $u \uparrow \mathfrak{c}_h$. Thus there is exactly one lobe maximum in every nodal interval, including the outer interval $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathfrak{c}_h)$. Let u_K^{max} be the unique maximizer on $\mathcal{J}_K^{\text{out}}$. For every $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$, there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathfrak{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If $z_K(\ell) \geq 1$, write it as $\cosh \eta$, $0 \leq \eta \leq h$; if it is at most 1, write it as $\cos \phi$, $0 \leq \phi < \pi/K$. These are two coordinates on the same outer lobe, joined at $z = 1$.

For a positive-half gap and $\theta \in (j\pi/K, (j+1)\pi/K)$, $\theta \leq \pi/2$, a j -chord is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)};$$

its signed root is $(-1)^j s$. The term endpoint-side always refers to the part of the outer lobe adjacent to $\mathfrak{c}_h$, not to the hyperbolic or elliptic coordinate used for it.

We now prove that these chords are exactly the middle signed branch. By reflection it suffices to consider a positive-half gap, so $j \geq 1$ and $\theta \leq \pi/2$. Put $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$. Since $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$ and

$$\mathcal{F}_h(t) = \frac{\sqrt{\mathfrak{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on $(0, \pi/2]$,

$$(4.28) \quad \Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u).$$

Thus the endpoint-side outer solution exists for every inner parameter.

At $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$, $s = 0$, the two folded variables are $\cos \frac{2k\pi}{K}$ and $\cosh(2h)$, or, in half variables, $\cos \frac{k\pi}{K}$ and $\mathfrak{c}_h$. Moreover, zero is a simple eigenvalue of $B_{K-1}(\lambda_{K-1,k})$: the matrix $\lambda_{K-1,k} I_{K-1} - A_{K-1}$ has rank $K-2$, and J_{K-1} is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables $y = \cos \theta$ and z .

Continue this ordered middle root $s_*(x)$ through the open gap. Write $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$ for its folded discriminant, let $y(x) \leq z(x)$ be the reconstructed non-negative half variables on the real sheet issuing from the endpoint germ, and set $\Theta(x) = \arccos y(x)$. Let $\mathcal{C}$ be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \Theta(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\max} < z(x) < \mathfrak{c}_h.$$

On $\mathcal{C}$, the reversible folded determinant identities give $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus $\mathcal{C}$ is precisely the selected endpoint-side chord sheet. The endpoint germ just constructed makes $\mathcal{C}$ nonempty. The ordered eigenvalues of the continuous real symmetric pencil $B_{K-1}(x)$ are continuous in x , including at an eigenvalue crossing, so a multiple least singular value does not interrupt the continuation. The folded pair $u = 2z^2 - 1$, $v = 2y^2 - 1$ and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus $\mathcal{C}$ is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. If the weak limit had $\Delta_*(x) = 0$, then its two reconstructed half variables would coincide. This is impossible because

$$z \geq u_K^{\max} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

The signs of the half variables cannot change in a noncentral positive gap: $z > 0$ throughout, while $y = 0$ would force $x = 0$, outside such a gap. Moreover, z cannot reach $\mathfrak{c}_h$ in the interior, because (4.25) would then give $s = 0$. The inner variable cannot reach a nodal endpoint: the common level would be zero, the endpoint-side solution would be $\mathfrak{c}_h$, and (4.25) would give $s = 0$, whereas the middle root has no zero inside a spectral gap. The outer variable cannot leave its endpoint-side component, since its first such point would be $u_K^{\max}$, contradicting (4.28). Passage through $z = 1$ is harmless; only its coordinate changes. Hence every relative closure point remains in $\mathcal{C}$. Since the open gap is an interval, $\mathcal{C}$ is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle Θ extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\Theta(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \Theta(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every $\theta_0 \in (j\pi/K, (j+1)\pi/K)$, the intermediate value theorem therefore supplies an interior gap point x with $\Theta(x) = \theta_0$. The classification just proved and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed j -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By Lemma 4.2, its positive magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$. Both exhaustion directions have now been proved without assuming that $\theta \mapsto x$ is monotone.

When K is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to $x = 0$. Because $z > 0$, the

identity $x = (2r/\mathfrak{c}_h) \cos \theta z$ forces $\theta = \pi/2$ at $x = 0$. Thus every parameter on that half occurs, and reflection covers the negative half.

Noncentral gap comparison. We now compare the noncentral middle branches. Let $N \geq 3$, put $M = N + 1$, and set $\kappa = N/M$. Every positive noncentral gap of the M -problem satisfies

$$(4.29) \quad j \geq 1, \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

Indeed, such gaps satisfy $j+1 \leq N/2$. Hence $\theta_0 = \theta/\kappa$ belongs to the same-indexed N -nodal interval, and every trigonometric argument below lies in $(0, \pi/2]$. Consider a point of such a branch for the matrix of order $M - 1$, with

$$(4.30) \quad \theta \in \left(\frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right), \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

First suppose $\Phi_M(\theta) = \Psi_M(\eta)$. Put $\theta_0 = \theta/\kappa$. Then θ_0 belongs to the corresponding N -nodal interval. Concavity of sine gives

$$(4.30) \quad \frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{F}_h(\theta_0)}{\mathcal{F}_h(\theta)} > \kappa.$$

Indeed, $\theta = \kappa\theta_0$ and strict concavity gives $\sin \theta > \kappa \sin \theta_0$; hence

$$(4.31) \quad \mathcal{F}_h(\theta_0)^2 - \kappa^2 \mathcal{F}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left(\frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$(4.31) \quad \frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa,$$

with strict inequality for $\eta > 0$, by convexity of $\sinh$ on the positive half-axis. Hence $\Phi_N(\theta_0) > \Psi_N(\eta)$. Since Φ_N vanishes at the right endpoint $(j+1)\pi/N$, its rightmost solution $\widehat{\theta}$ of

$$\Phi_N(\widehat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies $\widehat{\theta} > \theta_0 > \theta$. Strict log-concavity of $\mathcal{W}_N$ makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the N -problem as well. Indeed, for $u = \cosh \eta > 1$,

$$(4.32) \quad \frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for $\eta > 0$,

$$(4.33) \quad \frac{d}{dt} (t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the N -derivative strictly negative. The case $\eta = 0$ follows by continuity. The chord classification above says that the constructed value

542 $\mathfrak{s}_{\text{hyp}}(\widehat{\theta}, \eta)$ is the least singular value at its constructed abscissa. Formula (4.25) and
 543 the strict increase of sine on the interval now give

$$544 \quad (4.32) \quad \mathfrak{s}_{\text{hyp}}(\widehat{\theta}, \eta) > \mathfrak{s}_{\text{hyp}}(\theta, \eta).$$

545 Suppose next that the exterior endpoint is elliptic, so that $\Phi_M(\theta) = \Phi_M(\phi)$,
 546 $0 < \phi < \pi/M$. (The case $\phi = 0$ is the limiting hyperbolic case already covered.) Put

$$547 \quad \theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

548 The ratio

$$549 \quad \mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{F}_h(t/\kappa)}{\mathcal{F}_h(t)}$$

550 is strictly increasing while $0 < t < t/\kappa \leq \pi/2$. Indeed,

$$551 \quad \frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

552 because

$$553 \quad t \mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

554 is strictly decreasing: both positive factors decrease on $(0, \pi/2)$, and the second
 555 decreases strictly. Since $\theta > \pi/M > \phi$, it follows that $\Phi_N(\theta_0) > \Phi_N(\phi_0)$. The
 556 rightmost solution of

$$557 \quad \Phi_N(\widehat{\theta}) = \Phi_N(\phi_0)$$

558 therefore has $\widehat{\theta} > \theta_0$. Its exterior variable is endpoint-side. Indeed, if $\Lambda_K(t) =$
 559 $(\log \Phi_K)'(t)$, then on the endpoint-side part of the first outer lobe $\Lambda_M(\phi) \geq 0$, and
 560 differentiation of $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$ gives

$$561 \quad \kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

562 Because $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$, positivity of $\Lambda_N(\phi_0)$ is exactly the endpoint-
 563 side condition for the N -outer solution. The one-maximum property now fixes the
 564 label, and the chord classification says again that $\mathfrak{s}_{\text{ell}}(\widehat{\theta}, \phi_0)$ is the least singular value
 565 at the constructed abscissa. Here $\widehat{\theta} > \theta_0 > \theta$ and $\phi_0 > \phi$, with all four angles below
 566 $\pi/2$. Thus both sine factors in the elliptic version of (4.25) have increased, at least
 567 one strictly, and hence

$$568 \quad (4.33) \quad \mathfrak{s}_{\text{ell}}(\widehat{\theta}, \phi_0) > \mathfrak{s}_{\text{ell}}(\theta, \phi).$$

569 *Central gaps.* It remains to compare the central sheets. We do this in the next
 570 calculation.

571 For $m \geq 1$, let

$$572 \quad (4.34) \quad c_m = s_{\min}(aI_m + V_m).$$

573 By reflection, the two gaps adjacent to the zero eigenvalue of A_{2m+1} have the same
 574 height; denote it by

$$575 \quad (4.35) \quad d_m = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}).$$

576 We claim

$$577 \quad (4.36) \quad c_m > d_m > c_{m+1}.$$

578 Put

$$579 \quad \zeta = \frac{a + a^{-1}}{2}.$$

580 For every $j \geq 1$, there is a unique $\rho_j > \cos \frac{\pi}{j+1}$ such that

$$581 \quad (4.37) \quad aU_j(\rho_j) = U_{j-1}(\rho_j).$$

582 Indeed, the Christoffel–Darboux identity gives, above the largest zero of U_j ,

$$583 \quad \frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

584 The ratio decreases from $+\infty$ to 0, proving existence and uniqueness. A three-term
585 determinant recurrence gives

$$586 \quad (4.38) \quad \det((aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j) = a^j (U_j(t) - a^{-1}U_{j-1}(t)).$$

587 Thus the largest root ρ_j gives the least eigenvalue. Together with

$$588 \quad U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

589 equations (4.37) and (4.38) give

$$590 \quad (4.39) \quad c_j^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}.$$

591 We shall also use that c_m is exactly the height of the central gap of A_{2m} . A
592 permutation separating odd and even coordinates gives

$$593 \quad A_{2m} \sim \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

594 Moreover,

$$595 \quad (I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

596 The singular values of the block off-diagonal matrix are the union of the singular values
597 of its two off-diagonal blocks. The Gram inequality shows that $s_{\min}(I_m + aV_m^T) \geq$
598 $s_{\min}(aI_m + V_m) = c_m$, and hence $s_{\min}(A_{2m}) = c_m$. To prove that this is the maximum
599 throughout the central gap, put $K = 2m + 1$ and use the half variables

$$600 \quad y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

601 On the positive half of the central middle branch, the preceding chord classification
602 gives

$$603 \quad (4.40) \quad \mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathfrak{c}_h}yz, \quad s^2 = \frac{4a}{\mathfrak{c}_h^2}(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2),$$

where $0 \leq y < \cos \frac{m\pi}{K}$ is in the central lobe and z is the endpoint-side outer solution. The function $\mathcal{W}_K$ is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathfrak{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (4.37) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (4.39), this yields

$$(4.41) \quad \mathcal{W}_K(z_0) = \mathfrak{c}_h, \quad 4a(\mathfrak{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = c_m^2.$$

At $x = 0$, Lemma 4.2 and the chord-exhaustion argument identify the nearest root with c_m , so z_0 is the endpoint-side outer solution of level $\mathfrak{c}_h$. If $x \neq 0$, the level in (4.40) is strictly below $\mathfrak{c}_h$, and hence its endpoint-side solution satisfies $z > z_0$. Equations (4.40)–(4.41) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = c_m^2.$$

Reflection handles the negative half of the gap. Thus its height is exactly c_m , attained only at $x = 0$.

We first prove $d_m < c_m$. Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

Here are the details of the central specialization. Take $K = 2m + 2$ and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap, $-1 < t_{\text{in}} < -\omega_m^e$, while the endpoint-side outer variable satisfies $\omega_m^e < t_{\text{out}} < \zeta$, where $\zeta = 2\mathfrak{c}_h^2 - 1$. The identity $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$ gives

$$\mathcal{W}_{2m+2}(y)^2 = (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \quad \mathcal{W}_{2m+2}(z)^2 = (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2.$$

Because $\zeta + 1 = 2\mathfrak{c}_h^2$, the formulas $x = (2r/\mathfrak{c}_h)yz$ and $s^2 = (4a/\mathfrak{c}_h^2)(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)$ become the following chord parametrization:

$$(4.42) \quad \begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) &= |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 &= \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned}$$

where t_{out} is the descending, or largest, solution. To see this from (4.24), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through $t_{\text{out}} = 1$ uses the elliptic coordinate $z = \cos \phi$.

Let $\rho = \rho_m$. The function $\log \mathcal{V}_m$ is strictly concave on (ω_m^e, ζ) . If $(\log \mathcal{V}_m)'(\rho) \geq 0$, the descending root satisfies $t_{\text{out}} \geq \rho$, because ρ lies at or before the unique

lobe maximum and the descending root lies at or after it. Suppose instead that $(\log \mathcal{V}_m)'(\rho) < 0$. Then $\rho > a$. Indeed, otherwise $\rho \leq a < 1$, so $\rho < 1$ and in particular $\rho^2 - 1 \neq 0$. The identity

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1-\rho^2)} - \frac{1}{2(\zeta-\rho)} \\ &\geq \frac{1}{2(1-\rho)} - \frac{1}{2(\zeta-\rho)} \geq 0, \end{aligned}$$

a contradiction. Thus the displayed quotient has been used only on its valid domain; if $\rho = 1$, the desired conclusion $\rho > a$ is immediate from $a < 1$ and no quotient is invoked. Writing $t_{\text{in}} = -\cos \phi_c$, $0 < \phi_c < \pi/(m+1)$, gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1+a}{2a} < \frac{1+\rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now ρ is on the descending side and $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$, so the unique descending root again satisfies $t_{\text{out}} > \rho$. In either case, $t_{\text{out}} \geq \rho$, and $t_{\text{in}} > -1$ makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = c_m^2.$$

This holds along the whole central sheet, so $d_m < c_m$.

We now prove $d_m > c_{m+1}$. Put $L = m+1 \geq 2$, and abbreviate

$$\rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L,$$

$$(4.43) \quad \mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{2L}.$$

Thus $a = u_-/u_+$ and $c_L = 1/u_+$. We first establish the scalar inequality

$$(4.44) \quad (1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L) \mathfrak{D}_L.$$

Put $b = u_+/u_- = 1/a$, and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives $u_-^2 Q_b = 1$, and

$$(4.45) \quad \mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b.$$

Indeed, $u_+ = bu_-$ and $u_-^2 = Q_b^{-1}$, so

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

and the formula for Δ_ξ follows from (4.43). The ratio U_L/U_{L-1} is strictly increasing above the largest zero of U_{L-1} , and equals 1 at $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Hence $\rho > \rho_{\text{ref}}$. Since $S_b - Q_b = 2b(\rho + \omega_L) > 0$,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore $f_L(x) = (1+x)U_{L-1}(x)$ is positive, strictly increasing, and strictly log-concave on $[\xi, \rho]$. In particular, $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$, and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b-\rho)-1}{\rho^2-1}, & \rho \neq 1, \\ \frac{2L^2+1}{6}, & \rho = 1. \end{cases}$$

The endpoint value follows directly from $U_{L-1}(1) = L$ and $U'_{L-1}(1) = L(L^2-1)/3$. Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1+2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since $f_L(\rho)^2 = (1+\rho)^2/Q_b$ and $\Delta_\xi = \Delta_\omega Q_b/S_b$, substitution into (4.44) reduces it to

$$(4.46) \quad \mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0.$$

We shall repeatedly use

$$(4.47) \quad \Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1.$$

Suppose first that $\rho > 1$. Put $u = b - \rho$ and $\Delta_+ = \rho^2 - 1$. Since $\mathfrak{d}_\rho > 0$, $u > 1/L$. Keeping ρ fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

This is strictly increasing in $u \geq 1/L$, and its value at $1/L$ is positive because $\Delta_\omega L < 1$. Thus $\mathcal{N}$ exceeds its value at $b = \rho + 1/L$. With $s_0 = \rho + \omega_L$ and $\omega_+ = 1 + \omega_L$, that value is

$$\Upsilon(s_0) = s_0 \left[\left(s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+(s_0 + \Delta_\omega)^2.$$

For $s_0 \geq \omega_+$,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (4.46) holds when $\rho > 1$. If $\rho = 1$, then $b = (L+1)/L$, $Q_b = L^{-2}$, and direct substitution, using the endpoint value of $\mathfrak{d}_\rho$, gives

$$\mathcal{N} = \Upsilon(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

692 The first term is positive by the preceding calculation, and every factor in the correction
 693 term is positive.

694 Suppose next that $\rho < 1$. Set

$$695 \quad v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

696 Writing $\rho = \cos \theta_\rho$ gives $0 < L\theta_\rho < \pi/2$ and

$$697 \quad u = b - \rho = \frac{\sin((L+1)\theta_\rho)}{\sin(L\theta_\rho)} - \cos \theta_\rho = \sin \theta_\rho \cot(L\theta_\rho).$$

698 Because $b > 1$, $u = \Delta_b + v > v > 0$. Also $\tan(L\theta_\rho) > L \tan \theta_\rho > L \sin \theta_\rho$, since
 699 $x \mapsto \tan x/x$ is strictly increasing on $(0, \pi/2)$. Hence

$$700 \quad (4.48) \quad 0 < v < u < 1/L.$$

701 With ρ fixed, write $\mathcal{N}_\rho^<(\cdot)$ for $\mathcal{N}$ viewed as a function of $\Delta_b = b - 1$. A direct expansion
 702 of (4.46) gives

$$703 \quad \mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

704 where

$$705 \quad \mathfrak{q}(u) = \Delta_- [2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega [u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

706 This is a concave quadratic. Moreover,

$$707 \quad \mathcal{N}_\rho^<(0) = \frac{v}{2-v} [(2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega)].$$

708 The elementary bounds, the second following from $\rho > \rho_{\text{ref}}$ and $1 - \rho_{\text{ref}} < \pi^2/[2(2L +$
 709 $1)^2] < 5/(2L + 1)^2$,

$$710 \quad (4.49) \quad \Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}.$$

711 The last inequality is an equality when $L = 2$; because the preceding bound for v
 712 is strict, one still has $v < 1/5$. These bounds show that the bracketed expression is
 713 positive, since it equals

$$714 \quad (2 - v) [(2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L] + 2\Delta_\omega^2(2L - 1)$$

715 and, using $L \geq 2$,

$$716 \quad (2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

717 At the endpoints of (4.48),

$$718 \quad \mathfrak{q}(v) = v [(2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v)] > 0,$$

$$719 \quad \mathfrak{q}(1/L) = v [(2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L] > 0.$$

721 Indeed, in these two lines the positive leading terms exceed

$$722 \quad \frac{9}{5} \left(4 - \frac{5}{8}\right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5}\right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than $(5/L)(6/5) \leq 3$ and $5/L \leq 5/2$, respectively. Concavity now gives $q(u) > 0$, proving (4.46), and hence (4.44), in all cases.

It remains to certify that the scalar comparison concerns the least singular value. Since $a > 0$, $\Delta_\omega > 0$, $\rho + \omega_L > 0$, and $\zeta + \omega_L > 0$, let $x_* > 0$ be the positive square root determined by

$$(4.50) \quad x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad c = c_L.$$

At $x = x_*$ and $s = \pm c$, the real pair $u = \xi$, $v = -\omega_L$ satisfies the two symmetric relations in (4.9). Indeed, its sum and product are $\xi - \omega_L$ and $-\omega_L\xi$, and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Moreover, $\xi + \omega_L > 0$, so the distinct branch of the divided difference in (4.20) applies. For convenience, write the folded numerator as

$$\mathfrak{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$X_\xi = x_*(\zeta - \xi), \quad Y_\xi = c(1 + \xi), \quad X_\omega = x_*(\zeta + \omega_L), \quad Y_\omega = c\Delta_\omega,$$

and put $U_\xi = U_{L-1}(\xi)$, $U_{-\omega} = U_{L-1}(-\omega_L)$. The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give $Y_\omega Y_\xi = X_\omega X_\xi$. The identity $u_-^2 Q_b = 1$ gives $Q_b > 0$, and $S_b - Q_b = 2b(\rho + \omega_L) > 0$, so $\mathfrak{D}_L = S_b/Q_b > 1$. Moreover, $L \geq 2$ gives $\omega_L > 1/2$, while $\rho > \rho_{\text{ref}} > \omega_L$; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{X_\omega^2}{Y_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with $\mathfrak{r}_* = X_\xi/Y_\xi = Y_\omega/X_\omega \in (0, 1)$, the common nonzero factor in (4.20) is $a^{L-1}/(\xi + \omega_L)$, and the two numerator differences are

$$\begin{aligned} \mathfrak{n}_c(\xi) - \mathfrak{n}_c(-\omega_L) &= (1 + \mathfrak{r}_*)(X_\omega U_{-\omega} - Y_\xi U_\xi), \\ \mathfrak{n}_{-c}(\xi) - \mathfrak{n}_{-c}(-\omega_L) &= (1 - \mathfrak{r}_*)(X_\omega U_{-\omega} + Y_\xi U_\xi). \end{aligned}$$

Their product is positive by (4.44), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad X_\omega^2 U_{-\omega}^2 = \frac{c^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > Y_\xi^2 U_\xi^2.$$

Thus

$$(4.51) \quad q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0.$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal $1 + a^2$ and off-diagonal a . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim

$$(4.52) \quad x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}.$$

Indeed, recall $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Since $\rho > \rho_{\text{ref}}$ and U_{L-1} is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities $c = a/u_-$ and $c^2 = 2a(\zeta - \rho)$ imply $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$. Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then $\Delta_\omega = 2 \sin^2 \theta_L$ and $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$. Since all quantities are positive, the preceding quotient bound for x_*^2 gives

$$x_* < 2\sqrt{a} \sin \theta_L.$$

Likewise,

$$c^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

and hence

$$c < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

where the last inequality uses $0 < a < 1$, hence $a < \sqrt{a}$. It remains to prove

$$(4.53) \quad \sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L).$$

Now

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L.$$

Thus (4.53) follows from $\Delta_L \cos \theta_L > \theta_L^3$, or $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$. For $0 < \theta_L \leq \pi/8$, using $\pi^2 < 10$,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently $x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L}$. Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1 - a)^2 \geq 0,$$

which proves (4.52).

Let $\mu_1, \dots, \mu_{2L-1}$ be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1}) J_{2L-1}.$$

Right multiplication by J_{2L-1} preserves singular values, so the numbers $|\mu_i|$ are precisely the singular values of $x_* I_{2L-1} - A_{2L-1}$. Moreover,

$$q_{2L-1}(x_*, c) q_{2L-1}(x_*, -c) = \prod_{i=1}^{2L-1} (\mu_i^2 - c^2).$$

The positive product in (4.51) is nonzero and therefore says that the number of indices with $|\mu_i| < c$ is even. On the other hand, Weyl's singular-value perturbation inequality and (4.52) give, for the second-smallest singular value s_{next} ,

$$s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > c,$$

so at most one index can satisfy $|\mu_i| < c$. The number is thus zero, and nonvanishing of the product excludes equality. Hence $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > c_L$. Since $0 < x_* < 2r \sin \frac{\pi}{2L}$, the point lies strictly in the positive central gap, and $d_{L-1} > c_L$. This completes the proof of (4.36).

Completion of the comparison. We finish the index comparison. Put $M = N + 1$, so the larger and smaller matrices have orders $M - 1$ and $N - 1$, respectively. If $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$, then $\Sigma A_n \Sigma = -A_n$, so reflection reduces every noncentral comparison to a positive gap. By the interiority observation following the definition of the gap heights, choose $x_j \in G_{M-1,j}^\circ$ at which its height is attained. The hyperbolic or elliptic construction above produces a point $\hat{x}_j \in G_{N-1,j}$, and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

The reflected inequality handles the corresponding negative gap.

If $M = 2\ell_* + 1$, the remaining single central height of the larger, even-order matrix is c_{ℓ_*} , whereas each central height of the smaller odd-order matrix is $d_{\ell_*-1} > c_{\ell_*}$. If $M = 2\ell_* + 2$, the two central heights of the larger odd-order matrix equal $d_{\ell_*} < c_{\ell_*}$, where c_{ℓ_*} is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. Choose j_* with $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$. Then

$$\gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

Since $M - 1 = n + 1$ and $N - 1 = n$, this proves $\gamma_{n+1} < \gamma_n$. $\square$

5. Topology, quantitative bounds, and the transition scale. We begin with the standard component property that reduces the planar topology to the real spectral interval.

LEMMA 5.1. *Every connected component of $\sigma_\varepsilon(A_n)$ contains an eigenvalue of A_n ; this standard component property holds for arbitrary matrices [9].*

Proof. The estimate

$$s_{\min}(zI_n - A_n) \geq |z| - \|A_n\|_2$$

shows that the pseudospectrum is bounded. Let $\mathcal{C}$ be one of its components. If an eigenvalue λ belonged to $\overline{\mathcal{C}}$, continuity of the least singular value would give a ball about λ contained in the pseudospectrum. That ball meets $\mathcal{C}$, so maximality of connected components places λ in $\mathcal{C}$.

Suppose that $\mathcal{C}$ contains no eigenvalue. Its compact closure is then disjoint from the spectrum. On a neighbourhood of $\overline{\mathcal{C}}$, the function $u(z) = \|(zI_n - A_n)^{-1}\|_2$ is continuous and subharmonic; see [26]. Moreover, $u \leq \varepsilon^{-1}$ on $\partial\mathcal{C}$, since a strict inequality would put a neighbourhood of that boundary point in the same component. The maximum principle gives $u \leq \varepsilon^{-1}$ throughout $\mathcal{C}$, contrary to the definition of the pseudospectrum. $\square$

The diagonal gauge introduced in the introduction gives

$$A_n = \mathcal{G}_n(r)[r(V_n + V_n^T)]\mathcal{G}_n(r)^{-1},$$

and hence

$$(5.1) \quad \sigma(A_n) = \left\{ 2r \cos \frac{k\pi}{n+1} : 1 \leq k \leq n \right\}.$$

Put

$$\Omega = \sigma_\varepsilon(A_n), \quad \Omega_{\mathbb{R}} = \Omega \cap \mathbb{R} = \{x \in \mathbb{R} : g_n(x) < \varepsilon\}.$$

Because A_n is real, the least singular value is invariant under complex conjugation. Lemma 3.1 therefore implies that

$$\mathcal{H}_t(x + iy) = x + i(1-t)y, \quad 0 \leq t \leq 1,$$

maps Ω into itself. If $\mathcal{C}$ is a component of Ω , the path $t \mapsto \mathcal{H}_t(z)$ starts in $\mathcal{C}$ and hence remains there. Thus $\mathcal{H}_t$ retracts $\mathcal{C}$ onto $\mathcal{C} \cap \mathbb{R}$.

By Lemma 5.1, this real intersection is nonempty. It is connected, being the continuous image $\mathcal{H}_1(\mathcal{C})$, and hence lies in one interval component $\mathcal{K}$ of $\Omega_{\mathbb{R}}$. Conversely, $\mathcal{K}$ is a connected subset of Ω meeting $\mathcal{C}$, so $\mathcal{K} \subset \mathcal{C}$. Therefore $\mathcal{C} \cap \mathbb{R} = \mathcal{K}$, and the components of Ω correspond bijectively to the interval components of $\Omega_{\mathbb{R}}$. In particular, each $\mathcal{C}$ retracts onto an interval and is contractible.

If Ω is connected, its real intersection is an interval containing every eigenvalue, and therefore contains their convex hull I_n^{sp} . Conversely, if $I_n^{\text{sp}} \subset \Omega$, that interval belongs to one component and contains every eigenvalue; Lemma 5.1 rules out any other component. Compactness of I_n^{sp} now gives

$$(5.2) \quad \Omega \text{ is connected} \iff I_n^{\text{sp}} \subset \Omega \iff \max_{x \in I_n^{\text{sp}}} g_n(x) < \varepsilon \iff \gamma_n < \varepsilon.$$

The strict inequality reflects the open definition of the pseudospectrum; equality leaves at least one real bridge point outside.

We now give quantitative estimates. Put $N = n + 1$ and recall $\mathcal{L}_n(a)$ and $\mathcal{U}_n(a)$ from (2.3)–(2.2).

PROPOSITION 5.2 (Sharp-order gap bounds). *For every $n \geq 2$,*

$$(5.3) \quad \mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a).$$

In particular, both bounds have order $(n+1)a^{n/2}$, with constants depending only on a .

Proof. For $x = 2r \cos \theta \in I_n^{\text{sp}}$, set

$$\mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

All entries of $(xI_n - A_n)\mathbf{v}$, except the last, vanish, and the last one is $r^n \sin(N\theta)$. Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since $\pi/N \leq \theta \leq (N-1)\pi/N$, this proves the upper bound.

For the lower bound, put $p_\theta = re^{i\theta}$, $q_\theta = re^{-i\theta}$, so that $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$.

For $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$, put $u_0 = u_{n+1} = 0$, and define maps from $\mathbb{C}^n$ to $\mathbb{C}^{n+1}$ by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

For $1 \leq k \leq n$, direct multiplication gives

$$(\mathbf{D}_\theta^* \mathbf{C}_\theta \mathbf{u})_k = (\mathbf{C}_\theta \mathbf{u})_k - q_\theta (\mathbf{C}_\theta \mathbf{u})_{k-1} = (p_\theta + q_\theta) u_k - u_{k+1} - p_\theta q_\theta u_{k-1}.$$

Hence

$$(5.4) \quad xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta.$$

The two Gram matrices are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal $1 + r^2$ and off-diagonal $-r$. Its eigenvalues are $1 + r^2 - 2r \cos \frac{k\pi}{N}$, $1 \leq k \leq n$. Consequently, for every $\mathbf{u} \in \mathbb{C}^n$,

$$(5.5) \quad \begin{aligned} \|\mathbf{C}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2, \\ \|\mathbf{D}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2. \end{aligned}$$

Put

$$\mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = \mathcal{S}_N^{-1/2} (\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2} (1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$\mathbf{C}_\theta = Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

the matrices Q_C, Q_D are isometries onto these ranges. Two hyperplanes in $\mathbb{C}^{n+1}$ intersect in dimension $n-1$, so the cross-Gram matrix $Q_D^* Q_C$ has $n-1$ singular values equal to 1; the remaining one is the cosine of the remaining principal angle, $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$. The geometric sum gives

$$|\langle \mathbf{u}_D, \mathbf{u}_C \rangle| = \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j-(N-1))\theta} \right|$$

$$(5.6) \quad = \frac{r^n(1-r^2)}{1-r^{2N}} \left| \frac{\sin(N\theta)}{\sin\theta} \right| =: \chi_N(\theta).$$

Finally,

$$xI_n - A_n = (D_\theta^* D_\theta)^{1/2} Q_D^* Q_C (C_\theta^* C_\theta)^{1/2}.$$

Using $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$ and (5.5) gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting $\theta = 3\pi/(2N)$, which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$(5.7) \quad \mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}.$$

These follow from $\sin \frac{\pi}{N} \geq 2/N$, $1 + r^2 - 2r \cos \frac{\pi}{N} \geq (1-r)^2$, and $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$. This proves the asserted sharp order. $\square$

Proof of Theorem 2.1. By Proposition 5.2, $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$. Lemma 4.1 says that γ_n is strictly decreasing. Equation (5.2) therefore shows that, once the pseudospectrum is connected, it stays connected in every larger dimension. This proves $N_e = N_f$. Contractibility of every component was proved immediately before (5.2).

If $\mathcal{L}_n(a) \geq \varepsilon$, then $\gamma_n \geq \varepsilon$, so the pseudospectrum in dimension n is disconnected. If $\mathcal{U}_n(a) < \varepsilon$, then $\gamma_n < \varepsilon$, so it is connected. Strict decrease of γ_n now gives (2.4).

The scale in (2.4) can be displayed in closed form. Put

$$\Theta_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N \geq 2 : \Theta_m < t \text{ for every } m \geq N\}.$$

Equations (5.7) and $\mathcal{U}_n(a) \leq \Theta_n/2$ give

$$(5.8) \quad \boxed{\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon)}.$$

Indeed, (5.2) gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\}.$$

The lower estimate $C_{\text{lb}}(a)\Theta_m \leq \gamma_m$ forces $\Theta_m < \varepsilon/C_{\text{lb}}(a)$ throughout the tail beginning at N_c . Conversely, $\Theta_m < 2\varepsilon$ for every $m \geq \mathcal{T}_r(2\varepsilon)$, together with $\gamma_m \leq \Theta_m/2$, forces connectedness throughout that tail. Thus the polynomial factor $n+1$, as well as the exponential factor $a^{n/2}$, is sharp on both sides.

For completeness, let

$$\beta = -\log r = \frac{1}{2} \log(1/a).$$

For $0 < t < 3r^2 = \Theta_2$, let $x_t > 2$ be the solution on the decreasing branch of $(x+1)e^{-\beta x} = t$. This branch exists because

$$0 < \beta r t < 3\beta r^3 \leq e^{-1};$$

the last inequality is $ue^{-u} \leq e^{-1}$ with $u = 3\beta$. Moreover, $t < \Theta_2$ places the descending root strictly beyond $x = 2$. Putting $y_t = x_t + 1$, the equation becomes $(-\beta y_t)e^{-\beta y_t} = -\beta r t$, whence $y_t = -\beta^{-1}W_{-1}(-\beta r t)$. Therefore

$$(5.9) \quad \mathcal{T}_r(t) = \left\lfloor -\frac{1}{\beta}W_{-1}(-\beta r t) \right\rfloor,$$

where W_{-1} is the lower real branch of the Lambert W -function. Indeed, the continuous function $(x+1)r^x$ is unimodal and, after its descending root, is strictly below t . Hence the integer tail condition is $N > x_t$; its least solution is $\lfloor x_t \rfloor + 1 = \lfloor x_t + 1 \rfloor$, including when x_t is an integer.

The standard lower-branch expansion [11]

$$-W_{-1}(-u) = \log(1/u) + \log\log(1/u) + O(1), \quad u \downarrow 0,$$

combined with (5.8) proves, for each fixed $a \in (0, 1)$, as $\varepsilon \downarrow 0$,

$$(5.10) \quad N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log\log(1/\varepsilon)) + O_a(1).$$

The remainder is not asserted to be uniform as $a \uparrow 1$. Finally, for $|x| \leq r$,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus $s_{\min}(xI_2 - A_2) \leq a$. At $x = 0$, the singular values of $-A_2$ are 1 and a , so equality is attained and $\gamma_2 = a$. Hence, by (5.2),

$$(5.11) \quad N_c(a, \varepsilon) = 2 \iff \varepsilon > a.$$

This proves all assertions. $\square$

6. Numerical illustrations and reproducibility. The computations in this section illustrate the theorem but are not used in its proof. The complete script `make_hatano_nelson_figures.py`, supplied with the manuscript, uses NumPy, SciPy, and Matplotlib. It constructs $A_n(a)$ directly, computes Cartesian pseudospectral contours by singular value decomposition, and obtains gapwise estimates of γ_n by bounded scalar optimization. As a separate check, let M_h be the largest exact least singular value sampled on uniform real-gap grids of maximum spacing h . The 1-Lipschitz property of $x \mapsto s_{\min}(xI_n - A_n)$ gives the rigorous implication

$$M_h \leq \gamma_n \leq M_h + \frac{h}{2}$$

for exact samples. The supplied implementation applies this test to NumPy/SciPy float64 SVD values, but it does not enclose SVD or eigenvalue roundoff. Its topology labels are therefore reproducible floating-point checks, conditional on the computed samples, rather than validated interval certificates. They avoid reliance on pixel connectivity or the scalar optimizer and are unused in every proof.

The five panels of Figure 1 illustrate the computed classification: two mode-resolved cases, the first single-cloud case, and two subsequent single-cloud cases. This is the clean Hatano–Nelson realization of the $\tilde{N}$ crossover described schematically by

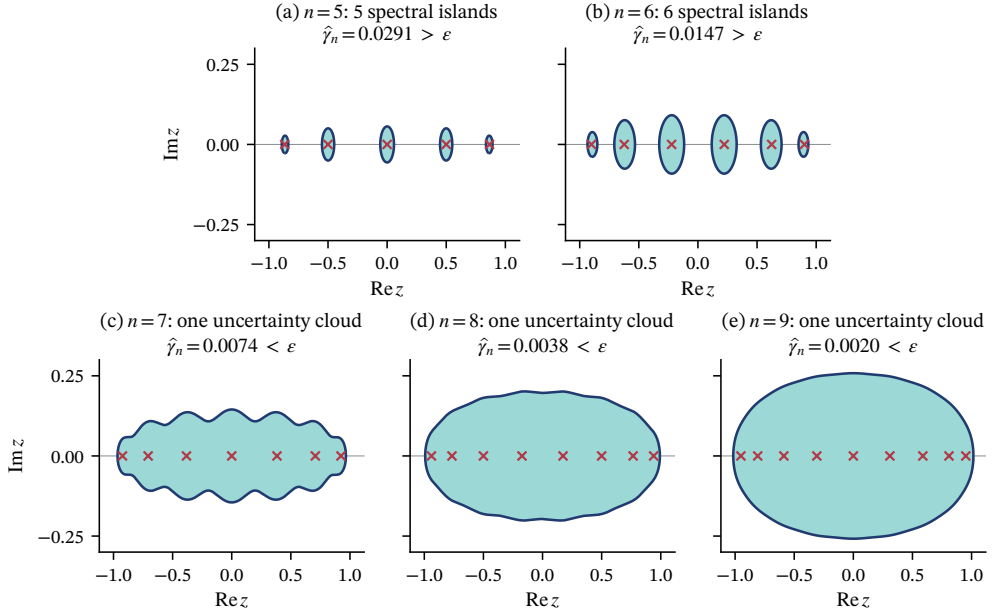


FIG. 1. Topology change indicated by the float64 computation at $a = 1/4$ and $\epsilon = 10^{-2}$. Shading depicts $\{z : s_{\min}(zI_n - A_n) < \epsilon\}$, and crosses are open-boundary eigenvalues. Gapwise SVD estimates give $(\hat{\gamma}_5, \dots, \hat{\gamma}_9) = (0.02909836, 0.01465829, 0.007364219, 0.003820033, 0.001992747)$. Applying the grid/Lipschitz check to the float64 samples places the computed margins for $n = 5, 6$ above ϵ and those for $n = 7, 8, 9$ below it, yielding the reported numerical threshold $N_c = 7$. As explained in the text, these checks are not interval certificates. In the two displayed pre-threshold cases every component contains one eigenvalue, so the components are mode-resolved uncertainty islands. At $n = 7$ the last separator is lost and all eigenvalues belong to one uncertainty cloud. The merger is a perturbation-level-dependent pseudospectral event, not the onset of skin localization.

Kiorpelidis and Makris [15]. As proved above, whenever the true pseudospectrum is connected it has no hole: every vertical section is an interval, and the real trace contains the full open-boundary spectral segment.

Figure 2 separates the two roles of the analysis. The strict comparison theorem identifies a unique crossing of a fixed ϵ , while the elementary upper and lower envelopes prove its sharp exponential-polynomial scale without numerically resolving the maximizing gap.

Figure 3 places the topological criterion and its nonnormal mechanism side by side. The left panels show the last real-axis separator being submerged below the fixed uncertainty floor; the right panel shows the imaginary-gauge envelope responsible for the exponentially small barrier scale. Skin localization already exists on both sides of the crossing.

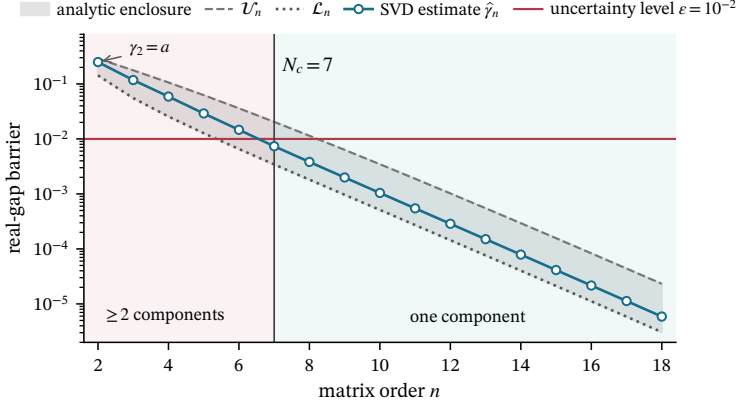


FIG. 2. Finite-size crossing of the perturbation budget. Gapwise float64 SVD estimates $\hat{\gamma}_n$ of the strictly decreasing barriers and the exact analytic enclosure $\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a)$ for $a = 1/4$. The plotted estimates first cross below $\varepsilon = 10^{-2}$ at $n = 7$, again indicating the reported numerical threshold. Independently, the exact analytic bounds in (2.4) give $6 \leq N_c \leq 9$. To the left of the crossing at least one component separator survives; from the crossing onward there is one component. The nearly parallel log-scale curves display the common $(n+1)a^{n/2}$ scale.

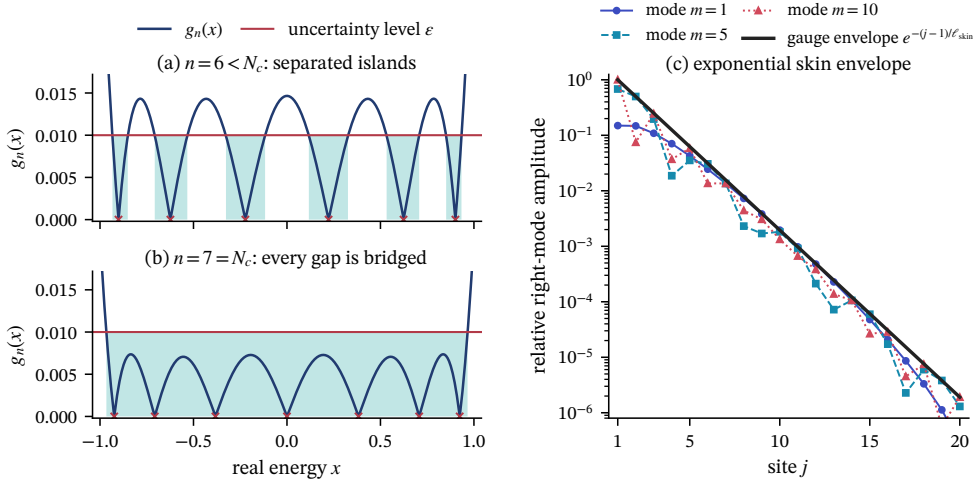


FIG. 3. Why the uncertainty clouds merge, for $a = 1/4$ and $\varepsilon = 10^{-2}$. In (a)–(b), the blue curves are $g_n(x) = s_{\min}(xI_n - A_n)$, the red line is the perturbation budget, and the teal portions are the real pseudospectrum. At $n = 6$, peaks above the budget separate the six eigenvalue islands; at $n = 7 = N_c$, every inter-eigenvalue location lies below the budget and the real trace is one interval. Panel (c) shows the common exponential envelope of right skin modes at $n = 20$. Its amplitude decay length is $\ell_{\text{skin}} = 2/\log(1/a)$, the coefficient in the asymptotic law for N_c . Reversing the site convention reverses the edge but not the merger criterion.

7. Hatano–Nelson interpretation: a finite spectral-resolution length. To connect the dimensionless matrix to the usual second-quantized notation, consider

$$\hat{H}_{\text{OBC}} = - \sum_{j=1}^{n-1} \left(t_L \hat{c}_j^\dagger \hat{c}_{j+1} + t_R \hat{c}_{j+1}^\dagger \hat{c}_j \right), \quad t_L > t_R > 0.$$

Its one-particle matrix is $-t_L A_n(a)$, $a = t_R/t_L$. The conventional minus sign is immaterial here because $\Sigma A_n(a) \Sigma = -A_n(a)$ for $\Sigma = \text{diag}(1, -1, \dots, (-1)^{n-1})$. Put

$$t_0 = \sqrt{t_L t_R}, \quad h = \frac{1}{2} \log(t_L/t_R) = \frac{1}{2} \log(1/a).$$

For the unitarily equivalent positive-sign representative $t_L A_n(a)$, multiplying the dimensionless gauge relation by t_L gives the eigenvalues and unnormalized right and left modes

$$\begin{aligned} E_m &= 2t_0 \cos \frac{m\pi}{n+1}, \\ \psi_m^R(j) &= e^{-h(j-1)} \sin \frac{j m \pi}{n+1}, \\ \psi_m^L(j) &= e^{h(j-1)} \sin \frac{j m \pi}{n+1}, \quad 1 \leq j, m \leq n. \end{aligned}$$

These mode profiles are closely related to the original Hatano–Nelson eigenfunction analysis [14]. Thus the right-mode amplitude skin depth is

$$\ell_{\text{skin}} = h^{-1} = \frac{2}{\log(1/a)},$$

whereas the gauge condition number is $\kappa_2(\mathcal{G}_n(e^{-h})) = e^{h(n-1)} = a^{-(n-1)/2}$.

7.1. Why a finite threshold exists. The least singular value has the backward-error interpretation

$$s_{\min}(zI_n - A_n) = \min\{\|\Delta A\|_2 : z \in \sigma(A_n + \Delta A)\}.$$

Consequently, γ_n is not merely an auxiliary quantity from the proof. It is the largest, over the real interval between the extreme open-boundary eigenvalues, of the minimum perturbation required to place an eigenvalue at the specified point. In other words, it is the cost of the last and most expensive bridge between spectral islands.

For the dimensional positive-sign Hamiltonian $H_n = t_L A_n(a)$, define the final-bridge budget

$$\varepsilon_*(n) = t_L \gamma_n.$$

Then the exact physical form of the connectedness criterion is

$$\sigma_{\varepsilon_{\text{phys}}}(H_n) \text{ is connected} \iff \varepsilon_{\text{phys}} > \varepsilon_*(n).$$

The strict decrease $\gamma_{n+1} < \gamma_n$, together with $\gamma_n \rightarrow 0$, explains why there is one finite threshold: a fixed horizontal uncertainty budget crosses the last-bridge scale once, and after that crossing the clouds cannot separate again. Equivalently, if $E_{\min}$ and $E_{\max}$ are the extreme eigenvalues of H_n , then

$$\varepsilon_*(n)^{-1} = \min_{x \in [E_{\min}, E_{\max}]} \|(xI_n - H_n)^{-1}\|_2,$$

with the resolvent norm interpreted as infinite at an eigenvalue. Thus at and beyond N_c , even the deepest inter-resonance valley has worst-case gain above $\varepsilon_{\text{phys}}^{-1}$.

The Hatano–Nelson content lies in the rate at which this bridge collapses. From Proposition 5.2,

$$\varepsilon_*(n) \asymp_a t_L(n+1)e^{-hn} = t_0(n+1)e^{-h(n-1)} = t_0(n+1)e^{-(n-1)/\ell_{\text{skin}}}.$$

The nonunitary imaginary gauge becomes exponentially ill-conditioned across the chain, and right and left skin modes accumulate at opposite edges. Accordingly,

$$\gamma_n^{-1} \asymp_a \frac{e^{hn}}{n+1},$$

so gauge conditioning and worst-case spectral sensitivity have the same exponential rate h , up to fixed- a constants and the displayed polynomial factor. Skin localization and barrier collapse are parallel consequences of the same imaginary gauge; the cloud merger is not the onset of the skin effect.

Let $\delta = \varepsilon_{\text{phys}}/t_L$. Balancing the fixed uncertainty level with the bridge scale above gives, as $\delta \downarrow 0$,

$$N_c(a, \delta) = \ell_{\text{skin}} (\log(1/\delta) + \log\log(1/\delta)) + O_a(1).$$

Thus N_c is a spectral-resolution length. Measured in amplitude skin depths, its leading value is $N_c/\ell_{\text{skin}} = \log(1/\delta) + \log\log(1/\delta) + O_a(1)$: this is the length needed for a fixed uncertainty floor to submerge the last robust gap. Improving the resolution by a factor q increases the leading resolvable length by $\ell_{\text{skin}} \log q$. Stronger nonreciprocity shortens ℓ_{skin} and hence lowers the leading small- δ threshold; this is an asymptotic statement, not a claim that N_c is monotone in a at every finite perturbation level.

The exact two-site endpoint makes the same interpretation especially transparent. Since $\gamma_2 = a$,

$$\varepsilon_*(2) = t_L a = t_R, \quad N_c(a, \varepsilon_{\text{phys}}/t_L) = 2 \quad \Longleftrightarrow \quad \varepsilon_{\text{phys}} > t_R.$$

Indeed, for the positive-sign dimer,

$$H_2 = \begin{bmatrix} 0 & t_L \\ t_R & 0 \end{bmatrix}, \quad \Delta H = -t_R \mathbf{e}_2 \mathbf{e}_1^T, \quad H_2 + \Delta H = \begin{bmatrix} 0 & t_L \\ 0 & 0 \end{bmatrix}.$$

The norm- t_R perturbation cancels the weaker directed hopping and produces a defective double eigenvalue at zero. Hence, if the uncertainty budget exceeds t_R , mode resolution is lost already in the shortest nontrivial chain; no long-chain buildup is needed. At exact equality the needed perturbation is excluded by the strict convention $\|\Delta H\|_2 < \varepsilon_{\text{phys}}$: the component closures touch at zero but the strict pseudospectrum is disconnected (indeed, $N_c(a, a) = 3$).

Kiorpelidis and Makris [15] denoted by $\tilde{N}$ the observed fixed-perturbation crossover from separate eigenvalue-centered clouds to a single cloud, and associated it with the end of exponential growth of the pseudospectral abscissa. For the clean open Hatano–Nelson chain, N_c is the exact finite-dimensional counterpart of that cloud-merger scale: the physical connectedness criterion above supplies the exact bridge criterion, and strict barrier decrease proves that the single cloud cannot split again at larger sizes. Their convention $A = -iH$ rotates the pseudospectrum by ninety degrees and does not change connectedness.

7.2. What is lost at the threshold. The relevant uncertainty is deterministic worst-case Hamiltonian or model uncertainty, not a quantum uncertainty principle. A point $z \in \sigma_\varepsilon(A_n)$ records only that *some* perturbation of norm below ε can place an eigenvalue at z ; the point does not come tagged with that perturbation or with a parent eigenvalue. Connected components nevertheless give a precise, perturbation-stable coarse label. Define two unperturbed eigenvalues to be in the same ε -resolution class when they lie in the same component of $\sigma_\varepsilon(A_n)$. For any continuous perturbation path $\Delta A(s)$ satisfying $\|\Delta A(s)\|_2 < \varepsilon$, every continuous eigenvalue branch of $A_n + \Delta A(s)$ remains inside $\sigma_\varepsilon(A_n)$ and therefore cannot pass between two distinct components. Equivalently, each component carries a fixed algebraic number of perturbed eigenvalues along such a path.

When a component contains one unperturbed eigenvalue, this class gives an individual-mode label. Partial mergers can combine some labels before the full threshold, so N_c need not be the first size at which one pair of eigenvalues becomes ambiguous. Rather, N_c is the *last* merger: for $n < N_c$ at least two robust spectral classes remain, whereas for $n \geq N_c$ the sole component contains all n eigenvalues. At that resolution there is no longer a perturbation-independent, topologically protected answer to “which open-boundary spectral cluster did this location come from?” If the actual perturbation or its continuation path is known, eigenvalue branches may still be tracked; connectedness says that the location alone no longer supplies the label.

This statement concerns a union over perturbations. It does not assert that one perturbation places eigenvalues at every point of a connecting bridge. Nor does it describe probabilities of observing different locations. In resolvent language, for $z \notin \sigma(A)$,

$$\|(zI_n - A)^{-1}\|_2 = \max_{\mathbf{f} \neq 0} \frac{\|\mathbf{u}\|_2}{\|\mathbf{f}\|_2}, \quad (zI_n - A)\mathbf{u} = \mathbf{f},$$

so the single cloud is a collective high-gain region at the specified worst-case resolution, rather than the spectrum of one noisy sample.

7.3. Scope and conclusion. The finite-size existence of a merger is not exclusive to the skin effect. At the normal endpoint $a = 1$, increasingly dense Hermitian eigenvalues also merge at fixed ε , but on the algebraic level-spacing scale. What is specifically Hatano–Nelson is the exponential collapse $(n+1)a^{n/2}$, the resulting logarithmic dependence of N_c on $1/\varepsilon$, and the appearance of ℓ_{skin} as its leading coefficient. This also explains why the limit $a \uparrow 1$ in the physical threshold asymptotic above is not uniform.

Restricting perturbations to onsite, hopping, boundary, or Toeplitz structure can produce a smaller attainable set and a different merger threshold, and requires a structured-pseudospectral analysis. Toeplitz-structured pseudospectra are treated in [7, 10]. Robustness under random imperfections is studied in [3]; pseudospectral boundary sensitivity in correlated or disordered systems in [20]; and complex and structured-pseudospectral scaling in asymmetric lattices in [15]. Likewise, the single-cloud threshold is a spectral-resolution crossover, not the end of dynamical exponential sensitivity; Kiorpelidis and Makris explicitly show that long-time dynamical amplification can persist beyond their $\tilde{N}$.

In summary, vertical monotonicity reduces planar connectedness to a backward-error barrier on the real axis, and strict barrier decrease makes its crossing with a fixed uncertainty budget unique. The imaginary gauge sets the exponentially small scale of that barrier, while the skin depth sets the leading finite-size resolution length. For the clean open Hatano–Nelson chain, N_c therefore marks the global loss of robust spectral-

cluster separation under unstructured worst-case uncertainty, without conflating that loss with the skin effect itself, a particular structured perturbation, or dynamical amplification.

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